

REPOSITORY HARDENING REPORT

512 / CVS Evidence-Sidecar / Constraint-Architecture

Multi-Session Hardening Pass — Execution Boundary Integrity, Vocabulary Lock, Schema Hardening, Layer Reference

Report period	17 April 2026 — 23 April 2026
Prepared by	Jonathan M. Watson
Pass type	Multi-session architectural hardening — execution boundary integrity, canonical IR correction, vocabulary lock, schema hardening, layer reference, constraint architecture methodology
Repos affected	3 — 512 (JonathanMastersWatson/512), CVS Evidence-Sidecar (JonathanMastersWatson/Evidence-Sidecar), Constraint-Architecture (JonathanMastersWatson/Constraint-Architecture)
Total files changed	40+ files across three repositories (new + modified)
Canon docs updated	512_ARCHITECTURE v2.9 → v3.0 512_IMPLEMENTATION v1.10 → v2.0 CVS_ARCHITECTURE v2.9 → v3.0 CVS_IMPLEMENTATION v2.4 → v2.5
Kernel	Unchanged — SHA-256: 7b08c024b77a24830c15e7952d6e54bed383aa960f4c74a71ff95ce51f4d80f5
Seal status	Pre-seal hardening phase. 512 and CVS repos marked. Constraint-Architecture repo is not sealed by design.

1. Executive Summary

This report documents the multi-session hardening pass conducted across the week of 17–23 April 2026, spanning three repositories: 512, CVS Evidence-Sidecar, and Constraint-Architecture. This is the most comprehensive hardening pass to date — closing all known semantic escape surfaces, establishing canonical vocabulary across all three repos, adding missing schema definitions, and introducing the upstream methodology that was documented in positioning but never operationalised.

The work was driven by two sources: (1) a post-hardening audit that identified specific contradictions and gaps remaining after prior passes, and (2) a structured hardening prompt identifying thirteen phases of work across binary output model enforcement, vocabulary lock, layer separation, schema hardening, historical containment, and conformance test hardening.

Three architectural issues of critical severity were identified and corrected:

- `CANONICAL_IR.json` contained 9 rules instead of 7 — misrepresenting the 7-statement kernel structure and producing an incorrect `SPEC_HASH` via `JCS(CANONICAL_IR.json)`
- The gate output model was still expressed as three values (`ALLOW/DENY/GAP`) in multiple files across both 512 and Constraint-Architecture repos
- CVS was described as 'optional infrastructure' in `512-core/CANON/README.md` — directly contradicting the full-system positioning established in all other docs

The canonical commitment hash (`7b08c024...`) is unaffected throughout. It is computed from `512-kernel-padded.txt`, not from `CANONICAL_IR.json`.

2. Sessions — Overview

Session	Date	Focus	Repos
1	17 Apr	GAP attribution correction — binary output model enforcement across 512 and CVS	512, CVS
2	22 Apr	Canonical IR correction (9→7 rules), execution boundary ambiguity in ops docs, CVS proof object terminology	512, CVS
3	22–23 Apr	Audit against hardening prompt — contradiction elimination, layer reference, schema hardening, vocabulary lock, historical containment	512, CVS
4	23 Apr	CVS equivalent pass — vocabulary lock, gate output matrix, layer reference, README de-versioning	CVS
5	23 Apr	Constraint-Architecture hardening — prohibited terms, wrong names, missing methodology, glossary expansion	CA
6	23 Apr	Canon docs updated — CVS_ARCHITECTURE v3.0, CVS_IMPLEMENTATION v2.5 downstream alignment	CVS

3. What Changed — by Repository

3.1 512 Repository

File	Action	What it addresses
512-core/CANON/CANONICAL_IR.json	Fixed	9-rule IR corrected to 7 rules. I3+I4→I3 (withdrawal+exit — one kernel statement). I8+I9→I7 (immutability+binary — one kernel statement). Rules renumbered. Ordering comment updated. SPEC_HASH now correctly represents 7-invariant kernel structure.
512-ops/COMMIT_BOUNDARY_REFERENCE.md	Modified	Binary output model hardened. §4 opening: 'exactly one of two outputs: ALLOW or DENY. No third value.' Fail-open moved to §4.1 as system behaviour, not gate output. Quick Reference table updated.
512-ops/REFERENCE_FLOW.md	Modified	Stage 4 hardened to two values only. Fail-Open moved to Stage 4.1. Observation point note updated: validation-result absent on fail-open; gap_record replaces it. All diagrams fenced.
FAILURE_MODES.md	Fixed	§3: 'one of three outputs: allow, deny, or gap' → 'exactly two outputs: ALLOW or DENY'. Last known instance of three-output model in 512-main. Fail-open path correctly described.
512-core/CANON/README.md	Fixed	'Optional infrastructure' framing for evidence sidecars replaced with three scoped statements: kernel-complete without witness layer; witness layer not optional in full deployment; this repo covers kernel only.
TERMS.md	Expanded	Eight new definitions added: Commit Boundary, Exclusive Commit Authority, Non-Bypassable Commit Path, Proposal Object, Compiled Constraint Set, Spec Hash, Gate Output (ALLOW/DENY only), Evidence Chain Gap, Witness Layer. Fail-Open rewritten.
LAYER_REFERENCE.md	New	Three-layer semantic firewall. Layer/Function/Must Not Do table. What Each Layer Produces table. What No Layer May Claim table (6 prohibited

		claims). Layer Separation Is Structural section. Core Invariant block.
README.md	Modified	LAYER_REFERENCE.md linked from Operational Reference table and Reference section.
512-ops/README.md	New	Operational directory index (was blank). Start Here table. Layer Reference section with core invariant. Key Principles for Builders. Related files.
512-ops/PROPOSAL_OBJECT.md	New	Single authoritative Proposal Object definition. 14 required fields with type, source, purpose, invariant dependency. Fields Required Before Evaluation. Construction Rules. Non-Conformant Patterns. Relationship to Evidence Objects.
512-ops/COMPILED_CONSTRAINT_FORMAT.md	New	Single authoritative compiled constraint definition. 7 required fields. Compiled Constraint Set requirements and loading sequence. Prohibited Content table. Complete I2 example in JSON.
ANTI_DRIFT.md	Modified	Prescribed vocabulary introduced in Single Authority Path, Pattern 7, Pattern 8: 'exclusive commit authority', 'non-bypassable commit path'. LAYER_REFERENCE.md and TERMS.md added to related docs.
CHANGELOG.md	Updated	Two new entries: [2026-04-22] Canonical IR Correction and Execution Boundary Hardening Pass; [2026-04-23] Contradiction Elimination and Layer Reference Pass; [2026-04-23] Schema Hardening, Vocabulary Lock, and Operational Reference Pass.

3.2 CVS Evidence-Sidecar Repository

File	Action	What it addresses
02_EVIDENCE_MODEL/PROOF_OBJECT_INTEGRITY.md	Fixed	Per-constraint 'gap' → 'unevaluated'. Distinguishing note added: per-constraint unevaluated (gate functioning, input missing) vs witness layer gap record (gate unavailable, no evaluation).
CONFORMANCE.md	Modified	Version-specific canon refs replaced with version-agnostic /08_CANON/ pattern. Test 2.1 qualified for fail-open events (validation-result absent; gap_record required). Test 5.1 precision update.
ANTI_DRIFT.md	Modified	'Exclusive commit authority' and 'non-bypassable commit path' introduced in §1.1. Fail-open behaviour described. §1.2 witness layer clarification. §2.1 fail-open observation path explicit. §15 conformance language fix. Related docs expanded and de-versioned.
README.md	Modified	Canon refs de-versioned to version-agnostic pattern. 'Conformance badge' language corrected. Reference section updated with UPSTREAM.md and GATE_OUTPUT_MATRIX.md entries.
02_EVIDENCE_MODEL/GATE_OUTPUT_MATRIX.md	New	Five-scenario matrix: Normal ALLOW, Normal DENY, Fail-open, Per-constraint unevaluated, Sidecar unavailable. Four columns: gate completed / per-constraint complete / gate result present / witness classification. Five critical semantic rules.
CHANGELOG.md	Updated	Two new entries: [2026-04-22] Proof Object Terminology and Conformance Hardening;

		[2026-04-23] Vocabulary Lock, Layer Reference, and Gate Output Matrix Pass.
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3.3 Canon Documents Delivered

Document	Version	Key changes
512_ARCHITECTURE	v2.9 → v3.0	New §2.5: Commit Gate Owns the Only Commit Path. Step 4 renamed: Execution Handoff → Commit Authorisation Signal. Commit path exclusivity in §8.1 Observable Properties and §8.2 Disqualifying Properties. Pre-hardening notice. Primitive Boundary section.
512_IMPLEMENTATION	v1.10 → v2.0	New §1.5: Commit Path Ownership Is Non-Bypassable. New §1.6: Non-Conformant Execution Patterns (5 patterns, ASCII diagrams). Step 4 renamed. Two new prohibited behaviours in §11.2. Two new verification tests in §11.4.
CVS_ARCHITECTURE	v2.9 → v3.0	Pre-hardening notice. New §0 Primitive Boundary subsection: CVS defines evidence formation at execution boundary; does not define system architecture or deployment models; derivative responsibility with implementing party.
CVS_IMPLEMENTATION	v2.4 → v2.5	Pre-hardening notice. New §0.7 Primitive Boundary. Cross-references updated from 512_ARCHITECTURE_v2.0 to v3.0 throughout.

3.4 Constraint-Architecture Repository

File	Action	What it addresses
README.md	Fixed	'binary — allow / deny / gap' → 'binary — ALLOW or DENY'. Fail-open path correctly described. Constraint Architect role description tightened: output is compiled constraint set, not policy document.
CANON/CVS.md	Fixed	'Cryptographic Verification System' → 'Cryptographic Verification Sidecar'. Canonical name corrected.
ROLES/512-integration-specialist.md	Renamed + Rewritten	Renamed from 512-compliant-consultant.md (filename typo + prohibited name). Role renamed from '512-Compliant Consultant' to '512 Integration Specialist'. Content rewritten: structural boundary work, exclusive commit authority, distinction from Constraint Architect.
GLOSSARY.md	Expanded	'Fail-fast' removed. Nine new definitions: Commit Boundary, Exclusive Commit Authority, Gate Output, Evidence Chain Gap, Compiled Constraint, Spec Hash, Proposal Object, Upstream, Downstream. Existing definitions tightened.
ROLES/cvs-systems-integrator.md	Fixed	'Cryptographic Verification Systems' → 'Sidecar'. '512-Compliant Consultant' → '512 Integration Specialist'. 'Compliance tooling' → 'conformance tooling'.
ROLES/constraint-architect.md	Modified	'Fail-fast' references replaced with fail-open model. Skill 3 rewritten as 'Fail-Open Design Thinking'. New Language list updated with canonical vocabulary. Conformance language corrected.
CONSTRAINT_TOOLS/constraint-definition-method.md	New	Upstream methodology document. Four-field definition model (Intent/Signal/Threshold/Authority). Directly-computable vs proxy invariant classification. Binary reducibility test. Anglo-Saxon test. Per-invariant definition checklist. Common failure modes including proxy drift.
PRE_HARDENING_NOTICE.md	New	Hardening phase status marker. Current pass record. Explicit statement that this repo is not sealed by design.

4. Issues Resolved

Complete issue register for the week's hardening passes, ordered by severity.

Issue	Severity	Where found	Resolution
CANONICAL_IR.json had 9 rules (should be 7)	Critical	512-main/512-core/CANON/CANONICAL_IR.json	Merged I3+I4→SPEC_HASH n kernel.
Gate output model: 'ALLOW/DENY/GAP' as three outputs	Critical	512-main: FAILURE_MODES.md, COMMIT_BOUNDARY_REFERENCE.md, REFERENCE_FLOW.md; CA: README.md	All instances co concept. Gate p
Fail-Open listed as third gate output sub-heading	Serious	COMMIT_BOUNDARY_REFERENCE.md §4, REFERENCE_FLOW.md Stage 4	Moved to §4.1 / section. Gate pr
CVS described as 'optional infrastructure'	Serious	512-core/CANON/README.md	Replaced with s without witness
'Cryptographic Verification System' — wrong CVS name	Serious	CANON/CVS.md (CA repo), cvs-systems-integrator.md	Corrected to 'Cr throughout.
Per-constraint 'gap' ambiguous vs witness layer gap	Serious	CVS/02_EVIDENCE_MODEL/PROOF_OBJECT_INTEGRITY.md	Replaced with 'u added.
'512-Compliant Consultant' — prohibited role name	Serious	CA/ROLES/512-compliant-consultant.md (also filename typo)	Renamed file ar
'Fail-fast' in glossary and mental models	Moderate	CA/GLOSSARY.md, CA/ROLES/constraint-architect.md	Removed from g Design Thinking
Version-specific canon refs in CVS operational docs	Moderate	CVS/CONFORMANCE.md, CVS/README.md	De-versioned to
Prescribed vocabulary missing from ops and anti-drift docs	Moderate	512/ANTI_DRIFT.md, CVS/ANTI_DRIFT.md, CA files	Introduced: excl bypassable com
No Proposal Object canonical definition	Moderate	512-ops/ (referenced but not defined as a single file)	512-ops/PROPO authoritative def
No Compiled Constraint canonical definition	Moderate	512-ops/ (referenced but not defined as a single file)	512-ops/COMP added.
No gate output / witness classification matrix	Moderate	CVS repo (gap between PROOF_OBJECT and CONFORMANCE)	02_EVIDENCE added — five-sc

No Constraint Architect upstream methodology	Moderate	CA repo (role described but work product process missing)	CONSTRAINT_ method.md add
Layer separation not documented as a standalone reference	Moderate	512 repo (scattered across ANTI_DRIFT, ops docs)	LAYER_REFER firewall docume
TERMS.md missing core operational vocabulary	Moderate	512-main/TERMS.md	Eight new defin operational voca

5. Vocabulary Lock — Cross-Repo Coverage

The following prescribed terms are now present across all three repositories. This table shows where each term appears.

Term	512	CVS	CA
exclusive commit authority	TERMS.md, LAYER_REFERENCE.md, 512-ops/README.md, ANTI_DRIFT.md	ANTI_DRIFT.md §1.1	GLOSSARY.md, 512-integration-specialist.md
non-bypassable commit path	TERMS.md, LAYER_REFERENCE.md, 512-ops/README.md, ANTI_DRIFT.md	ANTI_DRIFT.md §1.1	GLOSSARY.md
compiled constraint set	TERMS.md, COMPILED_CONSTRAINT_FORMAT.md, LAYER_REFERENCE.md	GATE_OUTPUT_MATRIX.md	GLOSSARY.md, constraint-definition-method.md
evidence chain gap	TERMS.md, LAYER_REFERENCE.md, ANTI_DRIFT.md	ANTI_DRIFT.md, GATE_OUTPUT_MATRIX.md, PROOF_OBJECT_INTEGRITY.md	GLOSSARY.md, constraint-architect.md
ALLOW / DENY only	TERMS.md, LAYER_REFERENCE.md, COMMIT_BOUNDARY_REFERENCE.md, REFERENCE_FLOW.md, ANTI_DRIFT.md, FAILURE_MODES.md	ANTI_DRIFT.md §1.1, CONFORMANCE.md, GATE_OUTPUT_MATRIX.md	README.md, GLOSSARY.md
unevaluated (per-constraint)	COMPILED_CONSTRAINT_FORMAT.md, PROPOSAL_OBJECT.md	PROOF_OBJECT_INTEGRITY.md, GATE_OUTPUT_MATRIX.md	—

6. Critical Test — Can the Repos Still Be Misread?

Can a developer implement a check-and-handoff model?

No. The non-conformant patterns (A through E) are explicitly diagrammed in COMMIT_BOUNDARY_REFERENCE.md §9 and 512_IMPLEMENTATION §1.6. ANTI_DRIFT.md Pattern 8 explicitly names pre-check positioning. LAYER_REFERENCE.md's prohibited claims table names it. 512-ops/README.md states 'exclusive commit authority' as a Key Principle. The term 'non-bypassable commit path' now appears in all three repos.

Can a vendor claim conformance while allowing bypass?

No. LAYER_REFERENCE.md's What No Layer May Claim table explicitly prohibits the claim. ANTI_DRIFT.md Pattern 7 states the disqualifying condition is existence of a bypass path, not frequency of use. PROPERTIES_CHECKLIST.md requires structural elimination of bypass paths. LAYER_REFERENCE.md states layer separation is structural, not procedural.

Can anyone still say the gate has three outputs?

No. The last known instance — FAILURE_MODES.md §3 — was corrected this pass. All other instances were corrected in the 17 April pass. TERMS.md, LAYER_REFERENCE.md, COMMIT_BOUNDARY_REFERENCE.md §4, REFERENCE_FLOW.md Stage 4, ANTI_DRIFT.md, GATE_OUTPUT_MATRIX.md, and CA/GLOSSARY.md all now state ALLOW or DENY only.

Can anyone say CVS is optional?

No. 512-core/CANON/README.md now carries three scoped statements: kernel-complete without witness layer; not optional in full 512/CVS deployment; this repo covers kernel only. LAYER_REFERENCE.md's prohibited claims table covers it. CA/README.md correctly describes CVS's role.

Can the Kernel/Boundary/Witness layers be conflated?

No. LAYER_REFERENCE.md provides the authoritative three-layer semantic firewall. It is linked from 512/README.md, 512-ops/README.md, CVS/ANTI_DRIFT.md, and CA/GLOSSARY.md. Each layer's Function and Must Not Do are stated in a single table a hostile reader cannot ignore.

7. New Files Added — Full List

File	Repo	Purpose
LAYER_REFERENCE.md	512	Three-layer semantic firewall — authoritative layer separation document
512-ops/README.md	512	Operational reference directory index with Key Principles for Builders
512-ops/PROPOSAL_OBJECT.md	512	Single authoritative Proposal Object definition — 14 fields, construction rules, non-conformant patterns
512-ops/COMPILED_CONSTRAINT_FORMAT.md	512	Single authoritative compiled constraint definition — 7 fields, set requirements, loading sequence
02_EVIDENCE_MODEL/GATE_OUTPUT_MATRIX.md	CVS	Five-scenario gate output and witness classification matrix with critical semantic rules
CONSTRAINT_TOOLS/constraint-definition-method.md	CA	Constraint Architect upstream methodology — four-field model, proxy vs direct, Anglo-Saxon test
PRE_HARDENING_NOTICE.md	CA	Hardening phase status marker with current pass record

8. What Remains Open

Item	Notes
CVS canon docs downstream alignment (CVS_ARCHITECTURE v3.0, CVS_IMPLEMENTATION v2.5)	Canon docs need downstream pass for 7-invariant IR correction and gate output model — not yet done
CA/ROLES/Career Transitions Into Constraint Architecture.md	References '512-Compliant Consultant' twice — needs updating to 512 Integration Specialist
CA/CANON/invariants.md — 'Execution is binary'	Confirm intentionally distinct from gate binary output model — may need scope note
CA README.md and CHANGELOG.md final versions	Deferred — to be done after all file changes complete
CANON_HASHES.md resealing	Held pending completion of full repo hardening pass
Async saga correlation mechanics	Explicitly out-of-scope frontier — not an omission
Multi-party constraint negotiation	Explicitly out-of-scope frontier — not an omission
Final CVS custody architecture patterns	Explicitly out-of-scope frontier — not an omission

9. Kernel Integrity Confirmation

Property	Value
Kernel text	7 statements — unchanged
Canonical artifact	512-core/KERNEL/512-kernel-padded.txt
SHA-256	7b08c024b77a24830c15e7952d6e54bed383aa960f4c74a71ff95ce51f4d80f5
XRPL genesis anchor	378536A3CB75DECF90B6AE57F75292BDFF716285B01946870CAC158F8152D100
Invariant count	7 — confirmed in kernel text, INVARIANTS.md, and now CANONICAL_IR.json
CANONICAL_IR rules	7 (I1–I7) — corrected this pass from 9 (I1–I9)
Canon docs hash	Not resealed — pre-seal hardening phase continues

10. Closing Statement

This week's pass addressed the most significant remaining semantic vulnerabilities across all three repositories. The binary output model is now uniformly enforced. The layer separation is documented in a single authoritative file. The prescribed vocabulary is present in all normative documents. The Constraint Architect's upstream methodology — previously described but never operationalised — now exists as a standalone document with a complete four-field definition model, directly-computable vs proxy invariant classification, and the Anglo-Saxon test.

The repos now form a coherent, cross-referenced architecture that is materially harder to misread, misrepresent, or build non-conformant implementations against. The critical test is met: a developer cannot implement a check-and-handoff model after reading these documents, and a vendor cannot claim conformance while allowing bypass.

The kernel is unchanged. The canonical commitment is intact. The hardening pass has not altered what 512 is — it has made what 512 is impossible to misstate.

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